Advanced Optical Fiber Communications

Problem 7: Measurement setup for RF power amplifier (dB calculation)

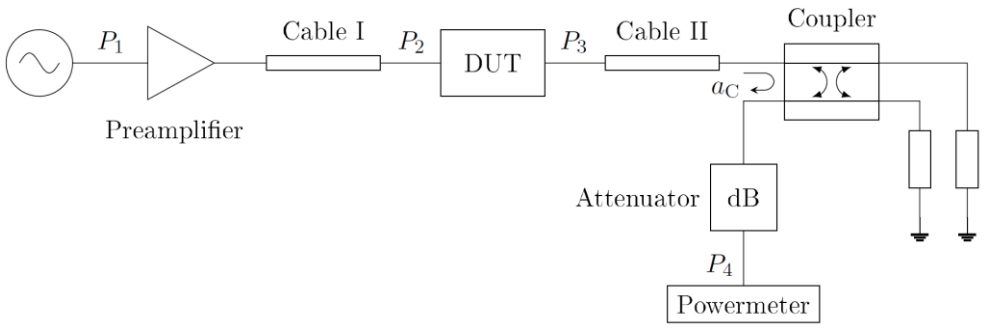


Figure 1 Schematic of the measurement setup

For measuring RF power amplifiers, a measurement setup based on fig. 1 is used. The following table gives the characteristics of the setup.

Max. power of the signal generator	$P_{sig,max}$ = 20 mW
Gain of the preamplifier	G_{pre} = 400
Max. input power of the preamplifier	$P_{1,max}$ = 16 mW
Coupling loss	a_C = 20 dB
Loss of the attenuator	a_A = 22 dB
Max. input power of the powermeter	$P_{4,max}$ = 45 mW
Gain of the device under test (DUT)	G_{DUT} = 10…40

a) Convert all powers from the table into dBm and all gains into dB.

In this setup, the cables and connectors cause unwanted attenuation, which need to be found with a calibration routine in order to make precise measurements of the DUT. The routine consists of two measurements, while the DUT is removed from the setup: At first, the power P_2 is measured with a known power P_1. Then, the nodes of P_2 and P_3 are directly connected and P_4 is evaluated.

b) Measurement 1: The power of the signal generator is $P_{sig} = P_1 = 5$ µW. The measured power is $P_2 = 1.8$ mW. What is the cable loss of cable I a_I?

c) Measurement 2: The measurement results in $P_4 = 100$ nW. Find the loss a_{II} of cable II.

Now, a power amplifier as DUT is being measured.

d) The input power at the DUT has to be $P_2 = 1$ W. What is the power at the signal generator (in dBm and in mW) that has to be fed to the preamplifier considering the calibration?

e) The power meter measures $P_4 = 0.8$ dBm. What is the output power P_3 of the DUT? Find the gain G_{DUT} in dB. How much is the power P_2 amplified by the DUT?

Problem 8: Planar waveguide

A planar waveguide consists of three different layers with different refractive indices. The layer in the middle has a thickness t and is expanded infinitely in the other directions, while the neighboring layers reach out infinitely in all directions based from the boundary.

a) Calculate the field profiles and find the characteristic equation for the effective refractive index for the TE polarization.

b) The thickness t is now 0.25 µm, the wavelength 1.55 µm, the refractive index of the middle layer is $n_2 = 3.47$, and the refractive index of the surrounding layers is $n_1 = n_3 = 1.44$ making the waveguide symmetric. What is the effective refractive index of the fundamental TE mode?

Problem 9: Step index fiber

A step index fiber shall be used in an optical communications channel. The refractive index for the core is $n_1 = 1.457$ and for the cladding $n_2 = 1.444$. The radius of the core $a = 6.1$ µm and the used wavelength is 1.55 µm.

a) Find the effective refractive indices for all modes that the step index fiber supports with the help of the B-V diagram. What are the corresponding incident angels of those modes at the core-cladding interface?

b) Calculate the waveguide dispersion group delay for all supported modes by using the d(VB)/dV diagram. Which mode is the slowest and which one the fastest?

The length of the used fiber is 2.3 km. Only the fastest and the slowest mode have to be considered from now on.

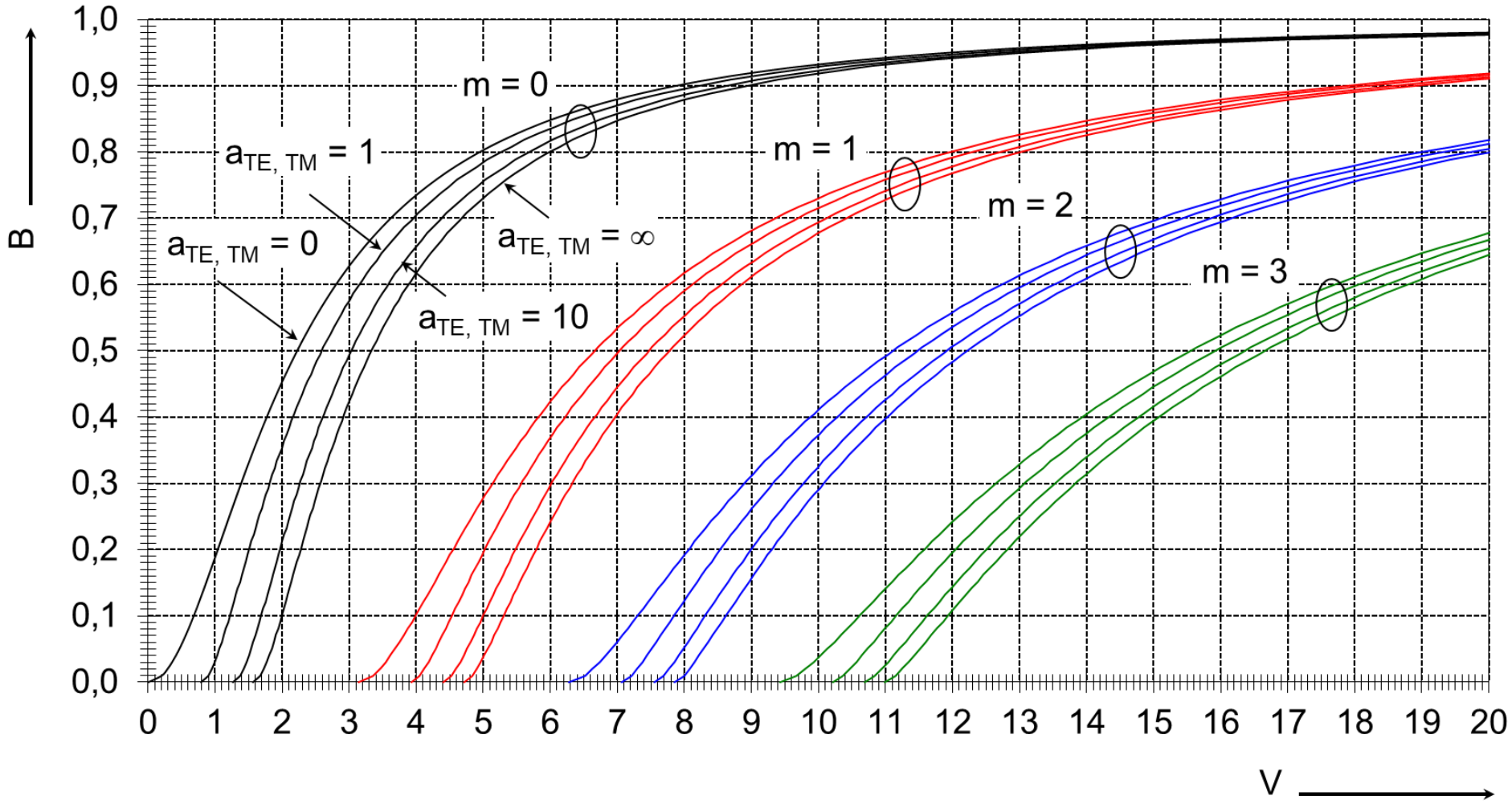
c) What is the time delay between the modes?

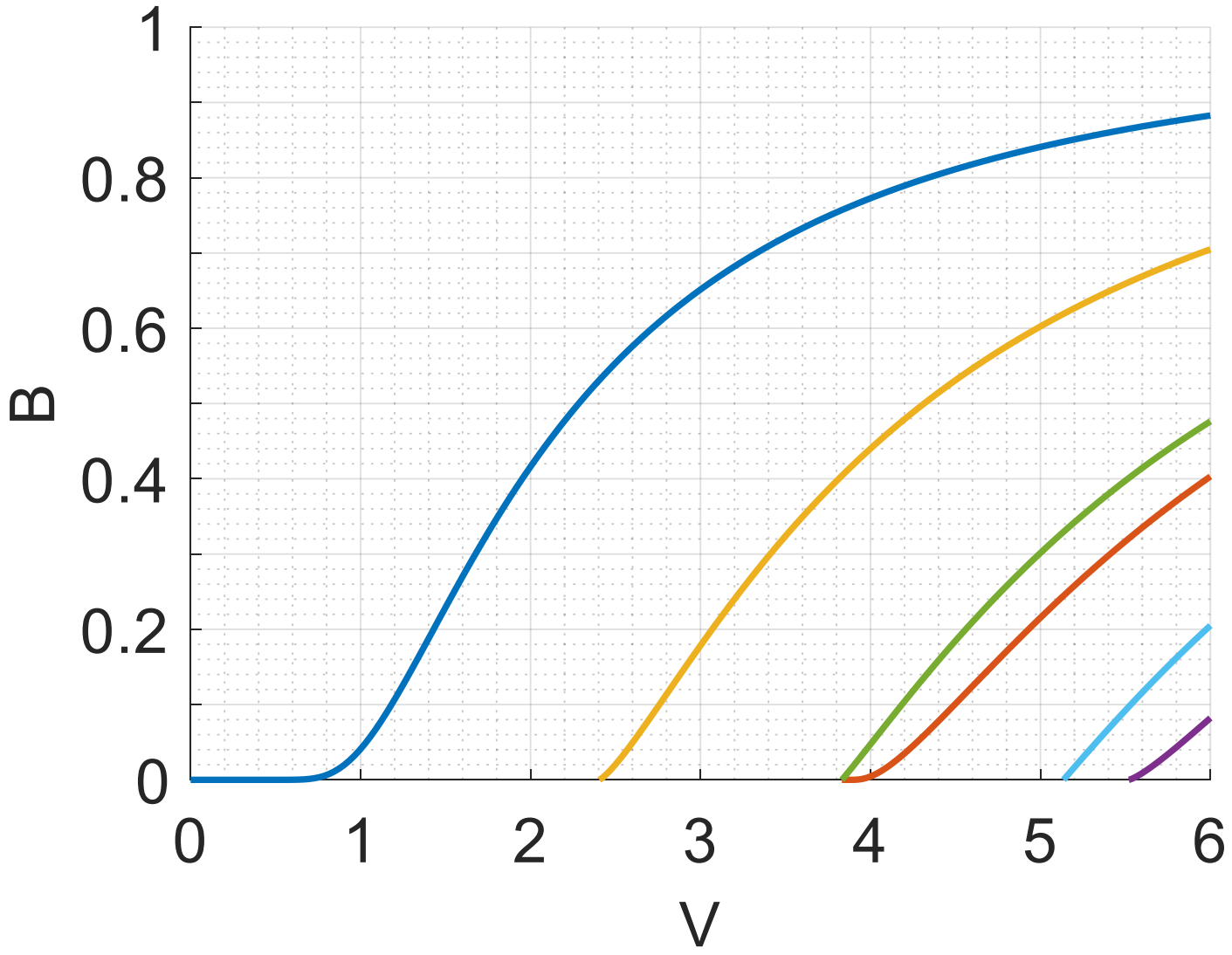
d) Sketch the signal at the output of the fiber for the symbol rates of 10 MBd and 30 MBd with the following NRZ coded symbols. The power is equally distributed on both modes.

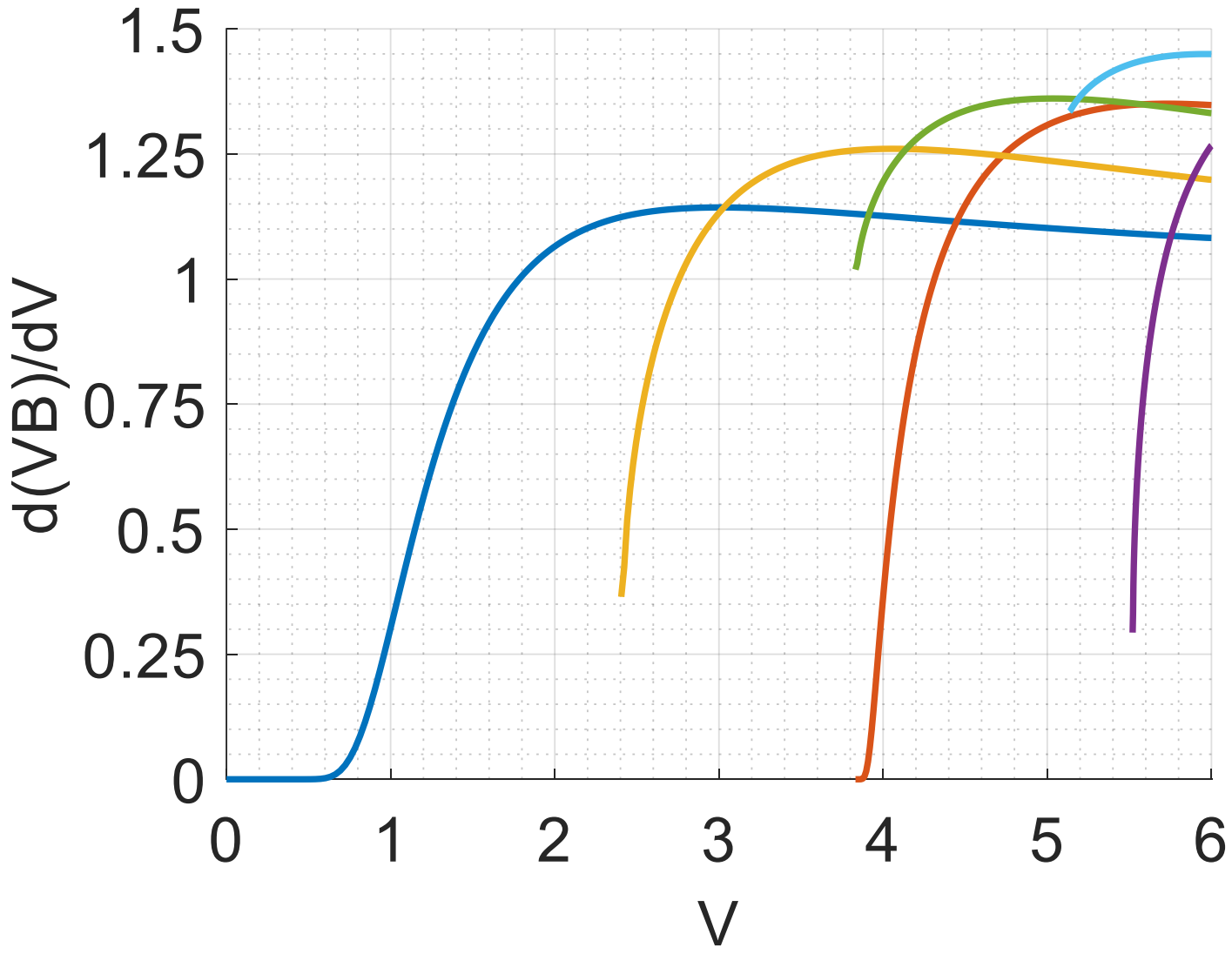
n	1	2	3	4	5	6	7	8	9	10
s_n	0	1	0	1	0	0	1	1	0	0

e) What is the bit error rate (BER) for both symbol rates, if the threshold is a hysteresis with the value 0.6 for 0-to-1 transition and 0.4 for 1-to-0 transition? The sampling takes place in the center of a symbol period.

B-V diagram planar waveguide




1
0.8
0.6
0.4
0.2
0
B
0
1
2
3
4
5
6
V


1.5
1.25
1
0.75
0.5
0.25
0
d(VB)/dV
0
1
2
3
4
5
6
V